

COL 761 Sem 1 2019 - 2020

Research Project

Note:

- You can use any language and package available on HPC

1 Problem Statement

Graphs are powerful and ubiquitous data structures that can model various kinds of real-world data. Modeling and generating graphs is fundamental for studying networks in biology, engineering, social sciences, etc.

You need to design and implement a graph generative model that learns the underlying distribution of a given set of graphs and generates similar graphs. This model should be domain-independent and scalable with the number of input graphs. Metrics for similarity measurement between set of graphs will be decided later in the course.

As a start, you can start by looking at [this](#) research paper.

Assumptions

1. Input graphs will have string labeled nodes and edges
2. Graphs will be undirected, connected and contain no self-loop

2 Input format

Graphs dataset will be provided in the form of a file where each graph will be in the following format

```
1  #graphID
2  No. of nodes
3  Series of node labels // one in each line
4  // Assume node indexing starting from 0 associated with them
5  No. of edges
6  Series of source node, destination node, edge label // one in each line
7  // Here source node and destination node are node indexes
```

Example [input file](#) containing graphs representing chemical molecules